Case inline: $x + \alpha + y$

Case fraction:

$$\frac{a+b}{c+\frac{d}{e}}$$

Case scripts:

$$x_{i_j}^{2^k} + y^2 + z_1 \tag{1}$$

Case radical:

$$u = \sqrt[3]{x + \sqrt{y}} \tag{2}$$

$$v = w \tag{3}$$

Case fence: $\left(\frac{x}{y}\right)$

Case matrix:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

Case sum:

$$\sum_{i=1}^{n} i + \mathbb{R} \tag{4}$$

Case cjk: $x + 中文 + y$

Case empty-frac:

Case empty-script: $x + y$

Case repeated: $x + x + \frac{x}{x} + x$

Case empty-matrix:

$$\begin{pmatrix} a & \\ & d \end{pmatrix}$$